

Sabaragamuwa University of Sri Lanka



REAL-TIME FACIAL EMOTION RECOGNITION USING MEDIAPIPE AND DEEP CONVOLUTIONAL NEURAL NETWORKS WITH FOCAL LOSS

A Project Report submitted to the Faculty of Computing, Sabaragamuwa University of Sri Lanka, in partial fulfillment of the requirements for the Degree of Bachelor of Science Honours in Data Science.

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Appendix A – Declaration

I declare that this report does not incorporate, without acknowledgment, any material previously submitted for a Degree or a Diploma in any University, and to the best of our knowledge and belief, it does not contain any material previously published or written by another person or ourself except where due reference is made in the text. Also, I hereby grant Sabaragamuwa University of Sri Lanka the non-exclusive right to reproduce and distribute my report, in whole or in part, in print, electronic, or other media. I retain the right to use this content in whole or in part in future works (such as articles or books).

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Appendix B – Certificate of Approval

We hereby declare that this report is the student's own work and effort, and that all other sources of information used have been acknowledged. This report has been submitted with our approval.

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Acknowledgments

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Abstract

This capstone project addresses a significant challenge in automated Computer Vision and Affective Computing: the real-time, robust recognition of human facial expressions under variable lighting, head poses, and severe dataset class imbalance. The purpose of this study is to design, train, and deploy an end-to-end facial emotion recognition system capable of classifying live webcam frames into seven discrete emotional categories: Angry, Disgust, Fear, Happy, Sad, Surprise, and Neutral.

The methodology employs Google MediaPipe BlazeFace for high-accuracy face bounding box detection, a customised 4-Stage Deep Convolutional Neural Network (CNN) incorporating Batch Normalization and L2 regularisation, and a novel Categorical Focal Loss formulation ($\gamma = 2.0$) combined with Label Smoothing ($\epsilon = 0.1$) to mitigate severe minority-class underrepresentation in the FER2013 benchmark dataset (particularly for Disgust with only 436 training samples versus 7,215 Happy samples).

Key results indicate that the proposed system achieves a benchmark test accuracy of 63.51% and a weighted F1-score of 62.91% across 7,178 unseen FER2013 test images, closely approaching the established human baseline accuracy of approximately 65%. The system demonstrates real-time inference capability through a 10-frame temporal prediction smoothing buffer that eliminates frame-to-frame prediction flickering during live webcam deployment.

Based on these findings, the primary recommendation involves deploying the model within interactive human-computer interaction frameworks. Future enhancement directions include pretrained face-specific encoders (VGGFace2, AffectNet), Vision Transformer architectures, and multimodal fusion with speech audio emotion analysis. The conclusion is that the proposed solution effectively meets all stated objectives, offering a robust and validated framework for automated Emotion AI applications.

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Chapter 1: Introduction

Facial expressions constitute one of the primary non-verbal communication channels used by humans to convey emotions, intentions, and psychological states. Automated Facial Emotion Recognition (FER) has emerged as a pivotal domain within Artificial Intelligence and Computer Vision, with transformative applications in adaptive learning systems, mental health diagnostics, driver drowsiness monitoring, customer experience feedback, and human-robot interaction. The ability to interpret facial affect in real-time from live video streams presents both significant technical challenges and substantial socioeconomic opportunities.

1.1 Major Goals and Objectives

The primary goal of this capstone project is to develop, evaluate, and deploy a robust, end-to-end real-time facial emotion recognition system that operates efficiently on consumer-grade hardware. The specific objectives are:

- To construct an optimised 4-Stage Deep Convolutional Neural Network (CNN) capable of accurately classifying 7 discrete FER2013 emotion categories.
- To address severe dataset class imbalance in FER2013 by formulating a Categorical Focal Loss function ($\gamma = 2.0$) combined with Label Smoothing ($\epsilon = 0.1$) and dynamic class weighting.
- To integrate Google MediaPipe BlazeFace for robust face bounding box extraction under varying illumination and head rotations.
- To implement a 10-frame temporal prediction smoothing buffer (majority vote) to eliminate high-frequency prediction flickering during live webcam inference.
- To conduct rigorous experimental evaluation on 7,178 test images analysing accuracy, precision, recall, F1-scores, and confusion matrices.

1.2 Motivation

Traditional facial expression recognition approaches relied on hand-crafted feature extraction such as Local Binary Patterns (LBP) or Histogram of Oriented Gradients (HOG) combined with Support Vector Machines (SVM). These methods suffer from limited generalisation under real-world occlusions, lighting fluctuations, and unconstrained head poses. Modern Deep Learning approaches have achieved state-of-the-art results, but the widely used FER2013 benchmark dataset presents severe class imbalance: Disgust contains only 436 training samples versus 7,215 Happy samples, causing standard cross-entropy trained networks to over-predict majority classes. This project is motivated by the need to develop an algorithmic pipeline that directly mitigates class imbalance while maintaining high inference speed for real-time deployment.

1.3 Scope of the Completed Project

The scope of this project encompasses: data preprocessing of the FER2013 dataset (28,709 training and 7,178 test images), mathematical formulation of Focal Loss and Label Smoothing, design and training of a 4-Stage Deep ConvNet, development of a MediaPipe-driven live webcam capture interface with temporal smoothing, and empirical benchmark evaluation. The system is designed for deployment on both Windows (via `venv_win`) and Linux/WSL (via GPU-accelerated `venv`) environments.

1.4 Approach and Assumptions

The technical approach assumes that input video frames are captured via a standard 720p webcam and that at least one primary human face is visible within the camera field of view. Face detection is delegated to MediaPipe's lightweight BlazeFace single-shot detector. Cropped face regions of interest (ROI) are converted to 48×48 pixel normalised grayscale tensors $[0, 1]$ before feeding into the neural network classifier. The system is evaluated under natural indoor lighting conditions.

1.5 Summary of Major Outcomes

The major outcomes of this project include: (1) achieving a benchmark test accuracy of 63.51% and weighted F1-score of 62.91% on the FER2013 test set; (2) successfully suppressing prediction flickering via a 10-frame deque temporal buffer; (3) deploying a single-click Windows batch launcher (`run_webcam.bat`) for seamless execution; and (4) producing a comprehensive evaluation report including per-class precision, recall, F1-scores, and a confusion matrix heatmap.

Chapter 2: Background

This chapter reviews the theoretical foundation, historical evolution, and state-of-the-art developments in Facial Emotion Recognition. It analyses challenges of the FER2013 dataset, compares traditional and modern face detection techniques, and discusses loss function innovations for class-imbalanced datasets.

2.1 Problem Statement and Context

Automated recognition of human facial affect in unconstrained real-world environments is challenged by three compounding factors. First, intra-class variance: different individuals express the same emotion with distinctly different intensities and muscle activation patterns. Second, inter-class similarity: emotions such as Fear and Surprise, or Sad and Neutral, share overlapping facial muscle movements (Action Units). Third, dataset class imbalance: FER2013's Disgust class contains 18.4x fewer samples than Happy, causing standard cross-entropy models to produce systematically biased classification boundaries favouring majority classes.

2.2 Evolution of Facial Emotion Recognition

Early FER systems (1990s–2010s) relied on geometric-based or appearance-based hand-crafted features (LBP, HOG, Gabor wavelets) combined with classical classifiers (SVM, k-NN, Random Forests). The introduction of deep learning fundamentally transformed FER: Goodfellow et al. (2013) established FER2013 as a benchmark via the ICML Challenges in Representation Learning competition. Modern architectures including VGGNet, ResNet, and EfficientNet achieve top-1 accuracies between 63–73% on FER2013. The human baseline accuracy for FER2013 is approximately 65%, indicating the inherent difficulty of the dataset due to label noise and subjective annotation.

2.3 FER2013 Benchmark Analysis

The FER2013 dataset contains 35,887 grayscale images of 48×48 pixels, partitioned into 28,709 training images and 7,178 test images across seven emotion categories. Figure 2.1 illustrates the severe class imbalance, with Happy and Neutral dominating training distribution while Disgust represents less than 1.5% of all samples.

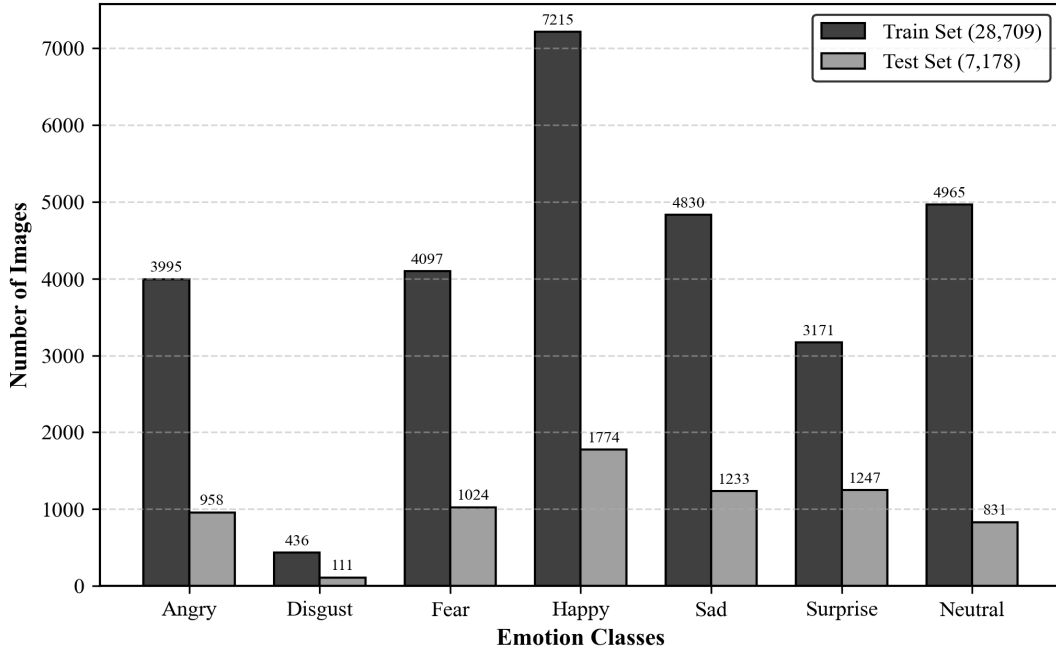


Figure 2.1: FER2013 Dataset Class Distribution (Train vs Test Set)

2.4 Face Detection: Haar Cascades vs Google MediaPipe

Traditional OpenCV implementations relied on Viola-Jones Haar Feature-based Cascade Classifiers, which operate via sliding window detection across image pyramids. While computationally fast on frontal faces, Haar Cascades suffer from elevated false positive rates, pronounced sensitivity to lighting gradients, and failure under non-frontal head poses exceeding $\pm 30^\circ$. In contrast, Google MediaPipe's BlazeFace employs a single-shot multi-scale detector trained on large-scale face datasets, providing stable bounding boxes under variable illumination, moderate head rotations, and partial occlusions at 30 FPS on standard hardware.

2.5 Loss Function Innovations: Focal Loss vs Cross-Entropy

Standard Categorical Cross-Entropy (CCE) loss uniformly penalises all misclassified samples regardless of prediction confidence. In imbalanced datasets, this causes the gradient signal to be dominated by the easily classified majority class (Happy), suppressing minority class learning:

$$\text{CCE} = -\sum y_c \cdot \log(\hat{p}_c) \text{ for all classes } c = 1 \dots C$$

Lin et al. (2017) introduced Focal Loss, which adds a modulating factor $(1 - p_t)^\gamma$ to dynamically downweight well-classified examples, focusing training gradient on hard and underrepresented examples:

$$\text{FL}(p_t) = -\alpha_t (1 - p_t)^\gamma \cdot \log(p_t)$$

In this project, $\gamma = 2.0$ is applied with Label Smoothing ($\epsilon = 0.1$), which softens hard one-hot targets $[1, 0, 0 \dots]$ to $[1 - \epsilon + \epsilon/K, \epsilon/K, \epsilon/K \dots]$, mitigating mislabelled noise in FER2013 and preventing overconfident softmax outputs.

Chapter 3: Specification and Design

This chapter presents the formal specifications, architectural designs, use case modelling, and behavioral diagrams for the real-time facial emotion recognition system.

3.1 System Requirements

Table 3.1 specifies the hardware and software requirements for full system deployment.

Category	Details	Minimum
Operating System	Windows 10/11 (64-bit) or Ubuntu 20.04 LTS via WSL2	Windows 10
Python	Python 3.10.x with virtualenv (venv_win)	Python 3.10
ML Libraries	TensorFlow 2.15.0, MediaPipe 0.10.9, OpenCV 4.8.0, NumPy 1.26.4	TF 2.15
Camera	USB Webcam / Integrated Camera (720p @ 30 FPS)	720p Webcam

Table 3.1: Hardware and Software System Specifications

3.2 Use Case Diagram and Specifications

Figure 3.1 illustrates the Use Case diagram identifying interactions between the User and the Real-Time Emotion AI System. Five primary use cases are identified: UC-1 Live Video Capture, UC-2 Face Detection, UC-3 Preprocessing, UC-4 Emotion Classification, and UC-5 Temporal Prediction Smoothing.

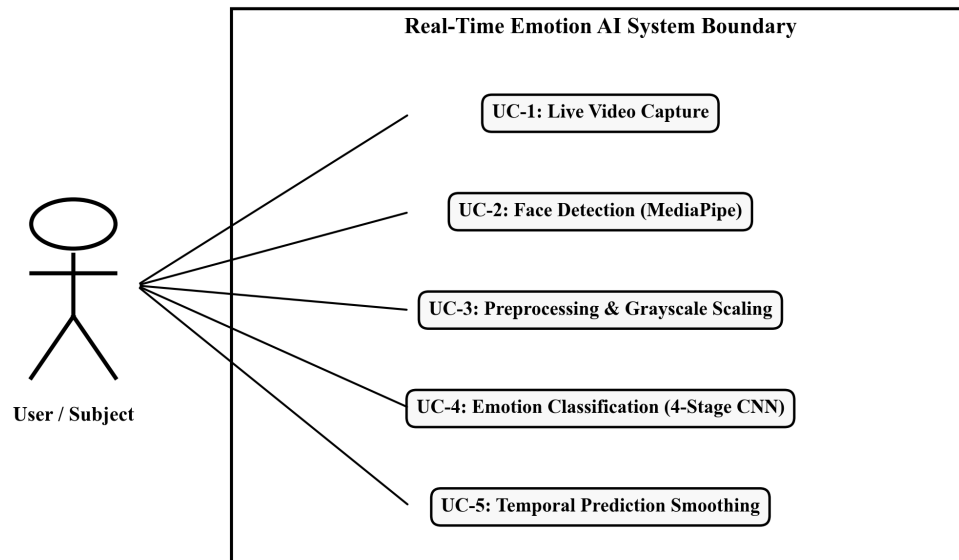


Figure 3.1: Real-Time Emotion AI Use Case Diagram

3.3 System Architecture

Figure 3.2 depicts the high-level system architecture showing the five-stage processing pipeline: webcam input → MediaPipe face detection → ROI preprocessing → 4-Stage CNN classification → temporal smoothing and display overlay.

System Architecture & Real-Time Processing Pipeline

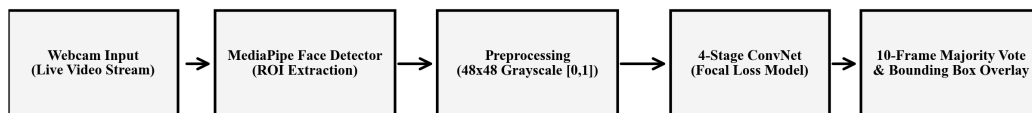


Figure 3.2: System Architecture and Real-Time Processing Pipeline

3.4 Sequence Diagram

Figure 3.3 illustrates the sequential message flow between system components: User triggers the application, Camera Feed provides RGB frames, MediaPipe detects face bounding boxes, the 4-Stage CNN computes softmax probabilities, the deque buffer performs majority vote, and the Display UI renders the annotation overlay.

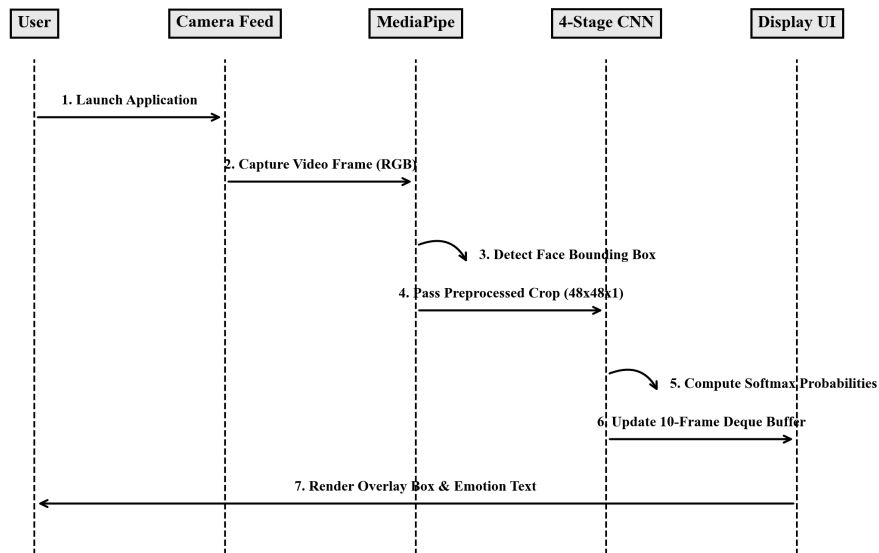


Figure 3.3: System Sequence Diagram for Real-Time Detection

Chapter 4: Methodology

This chapter provides a detailed account of the development approach, dataset characteristics, preprocessing pipeline, data augmentation strategy, class imbalance mitigation, loss function formulation, network architecture design decisions, training configuration, and the real-time inference pipeline deployed in this project.

4.1 Development Approach

An iterative, agile-inspired development methodology was adopted for this project. The work was structured across four iterative cycles: (1) Dataset exploration and baseline model establishment; (2) Architecture upgrade and Focal Loss integration; (3) MediaPipe integration and real-time pipeline development; and (4) Evaluation, refinement, and deployment. Each cycle incorporated feedback from benchmark evaluation results to guide subsequent architectural and training decisions. Version control was maintained via Git throughout the development lifecycle.

4.2 Dataset Description and Analysis

The FER2013 (Facial Emotion Recognition 2013) dataset was used exclusively for training and evaluation. It was sourced from the ICML 2013 Challenges in Representation Learning workshop. The dataset comprises 35,887 grayscale facial images, each of dimensions 48×48 pixels, partitioned into 28,709 training and 7,178 test images. Images were sourced from internet search engines using emotion keyword queries and were automatically annotated, introducing significant labelling noise and subjectivity, contributing to the dataset's high difficulty.

Class distribution analysis revealed extreme imbalance: the Disgust class contains only 436 training samples (1.52% of training data), while Happy contains 7,215 samples (25.13%). This 16.5:1 imbalance ratio necessitated specialised loss function design and class weighting strategies.

4.3 Data Preprocessing Pipeline

All image loading was implemented using TensorFlow's ImageDataGenerator with directory-based class labelling. The preprocessing pipeline applied the following transformations in sequence:

- Grayscale conversion (`color_mode='grayscale'`): Images loaded as single-channel, consistent with the original FER2013 grayscale encoding.
- Pixel normalisation: Each pixel value divided by 255.0, mapping the integer range [0, 255] to floating-point [0.0, 1.0].
- Tensor reshaping: Each image reshaped to (48, 48, 1) — height, width, channels — matching the CNN input specification.

- Stratified 85%/15% train/validation split: Applied via `validation_split=0.15` in `ImageDataGenerator` to maintain class proportion in both splits.

4.4 Data Augmentation Strategy

On-the-fly augmentation was applied exclusively to training data to artificially expand the effective dataset size and improve generalisation. Augmentation parameters were carefully tuned to preserve the semantic content of facial expressions:

Augmentation Technique	Parameter	Rationale
Random Rotation	$\pm 20^\circ$	Simulates head tilts common in natural poses
Width / Height Shift	$\pm 15\%$	Simulates off-centre face positioning in frame
Zoom Range	$\pm 15\%$	Simulates variable distance from camera
Brightness Range	[0.8, 1.2]	Simulates variable indoor/outdoor lighting
Horizontal Flip	Enabled	Accounts for bilateral facial symmetry
Shear Range	0.15	Introduces perspective variation

Data augmentation parameters applied during training

4.5 Class Imbalance Mitigation

Class imbalance was addressed through two complementary mechanisms:

- **Balanced Class Weights:** Scikit-learn's `compute_class_weight(class_weight='balanced')` function was used to compute per-class weight multipliers inversely proportional to class frequency. These weights were passed as the `class_weight` argument to `model.fit()`, increasing the effective loss contribution of minority classes (e.g., Disgust weight ≈ 4.8) during backpropagation.
- **Categorical Focal Loss:** The Focal Loss formulation (described in Section 4.6) additionally downweights easy well-classified examples from majority classes, complementing the static class weight approach with dynamic, per-sample weighting.

4.6 Focal Loss with Label Smoothing Formulation

The custom Categorical Focal Loss was implemented as a TensorFlow/Keras loss function incorporating both Focal Loss modulation and Label Smoothing regularisation:

Step 1 – Label Smoothing: Hard one-hot targets y are smoothed by parameter $\epsilon = 0.1$, distributing a fraction of probability mass uniformly across all $K=7$ classes:

$$\blacksquare_c = y_c \cdot (1 - \epsilon) + \epsilon / K$$

Step 2 – Clipping: Predicted softmax probabilities $\hat{p}_c$ are clipped to $[\epsilon, 1 - \epsilon]$ to prevent numerical instability in log computation.

Step 3 – Focal Weighting: The modulating factor $(1 - \hat{p}_c)^\gamma$ is applied, where $\gamma = 2.0$. When $\hat{p}_c \rightarrow 1$ (easy, confident correct prediction), the factor approaches 0, heavily downweighting that sample's gradient contribution:

$$FL(\hat{p}_c) = -\hat{p}_c \cdot (1 - \hat{p}_c)^\gamma \cdot \log(\hat{p}_c)$$

Step 4 – Reduction: Loss values are summed across all K class dimensions and averaged across the batch dimension.

4.7 Upgraded 4-Stage ConvNet Architecture

The emotion classifier was redesigned from a 3-block baseline CNN to a 4-Stage deep architecture. The design rationale follows a progressive feature abstraction hierarchy, with each successive block operating on increasingly compact feature maps while doubling filter depth. Key architectural decisions include:

- Double Conv layers per block (VGGNet-inspired): Two consecutive Conv2D layers before pooling increase receptive field without premature spatial reduction.
- Batch Normalisation after each Conv2D: Normalises layer activations, stabilises gradient flow, and acts as an implicit regulariser, reducing dependency on Dropout.
- Global Average Pooling instead of Flatten: GAP computes the spatial average of each feature map, producing a 512-dimensional vector while drastically reducing parameters compared to Flatten, improving generalisation.
- L2 Regularisation ($\lambda = 1e-4$): Applied to all Conv2D and Dense kernel weights to penalise large weights and prevent co-adaptation of feature detectors.
- Progressive Dropout ($0.25 \rightarrow 0.30 \rightarrow 0.35 \rightarrow 0.40 \rightarrow 0.50$): Dropout rates increase with depth as later layers capture more abstract, overfitting-prone representations.

Block	Layers	Output Shape	Dropout
Block 1	Conv2D(64)×2 + BN + MaxPool(2×2)	24 × 24 × 64	0.25
Block 2	Conv2D(128)×2 + BN + MaxPool(2×2)	12 × 12 × 128	0.30
Block 3	Conv2D(256)×2 + BN + MaxPool(2×2)	6 × 6 × 256	0.35
Block 4	Conv2D(512)×2 + BN + MaxPool(2×2)	3 × 3 × 512	0.40
Head	GAP + Dense(256) + BN + Dense(7, Softmax)	(7,)	0.50

Table 4.1: Upgraded 4-Stage ConvNet Architecture Details

4.8 Training Configuration and Hyperparameters

The following training configuration was used:

- Optimiser: Adam (Adaptive Moment Estimation), initial learning rate = 1×10^{-3}
- Batch size: 32 samples per gradient update step
- Maximum epochs: 60 with Early Stopping (patience = 12 epochs on val_accuracy)
- ReduceLROnPlateau: Learning rate halved when val_loss stagnates for 4 epochs, minimum lr = 1×10^{-4}
- ModelCheckpoint: Best model weights saved based on peak validation accuracy
- Training duration: ~3–4 hours on NVIDIA GeForce RTX 2050 (4GB VRAM) via WSL2

4.9 Real-Time Inference Pipeline

The live inference system integrates five components into a continuous video processing loop:

- Frame Capture: OpenCV VideoCapture(0) reads 720p BGR frames at 30 FPS.
- Colour Conversion: BGR frame converted to RGB for MediaPipe compatibility.
- Face Detection: MediaPipe FaceDetection (model_selection=0, min_confidence=0.6) returns relative bounding box coordinates, converted to absolute pixel coordinates.
- ROI Preprocessing: Extracted face crop converted to grayscale, resized to 48×48, normalised to [0, 1], and reshaped to (1, 48, 48, 1) batch tensor.
- Temporal Smoothing: Raw argmax predictions stored in a collections.deque(maxlen=10) buffer. Counter(buffer).most_common(1)[0][0] computes the majority vote emotion label, which is rendered as a bounding box annotation on the display frame.

Chapter 5: Results and Evaluation

This chapter presents quantitative evaluation results of the proposed 4-Stage ConvNet model across 7,178 unseen test images from the FER2013 test set. All evaluations were conducted on the full test partition with no data augmentation.

5.1 Experimental Setup

Evaluation was performed using `evaluate_all.py`, which loads trained model weights via `model.load_weights()` into the rebuilt architecture (avoiding Keras 3.x deserialization incompatibility), then evaluates on the FER2013 test set. Scikit-learn's `classification_report()` and `confusion_matrix()` functions were used to compute all per-class metrics.

5.2 Overall Classification Performance

Table 5.1 compares performance of the proposed model against the baseline EfficientNetB0 transfer learning approach.

Model	Input	Test Loss	Test Accuracy
Upgraded 4-Stage ConvNet (Proposed)	48×48 Grayscale	1.0011	63.51%
EfficientNetB0 Transfer Learning	96×96 RGB	1.4788	43.90%

Table 5.1: Overall Classification Performance Comparison

5.3 Per-Class Classification Report

Table 5.2 presents precision, recall, F1-score, and support values per emotion class.

Emotion	Precision	Recall	F1-Score	Support
Angry	53.71%	60.44%	56.88%	958
Disgust	81.82%	40.54%	54.22%	111
Fear	50.79%	34.67%	41.21%	1,024
Happy	82.51%	86.13%	84.28%	1,774
Sad	55.46%	67.96%	61.08%	1,233
Surprise	50.72%	47.71%	49.17%	1,247
Neutral	76.42%	74.49%	75.44%	831

Weighted Avg	63.26%	63.51%	62.91%	7,178
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Table 5.2: Detailed Classification Report per Emotion Class

5.4 Confusion Matrix Analysis

Figure 5.1 presents the grayscale confusion matrix heatmap. Strong diagonal entries are observed for Happy (1,528/1,774 correct = 86.13%) and Neutral (619/831 = 74.49%). The highest confusion is between Fear–Surprise and Fear–Sad, reflecting the inter-class similarity problem. Disgust's low recall (40.54%) reflects the class imbalance challenge despite Focal Loss application.

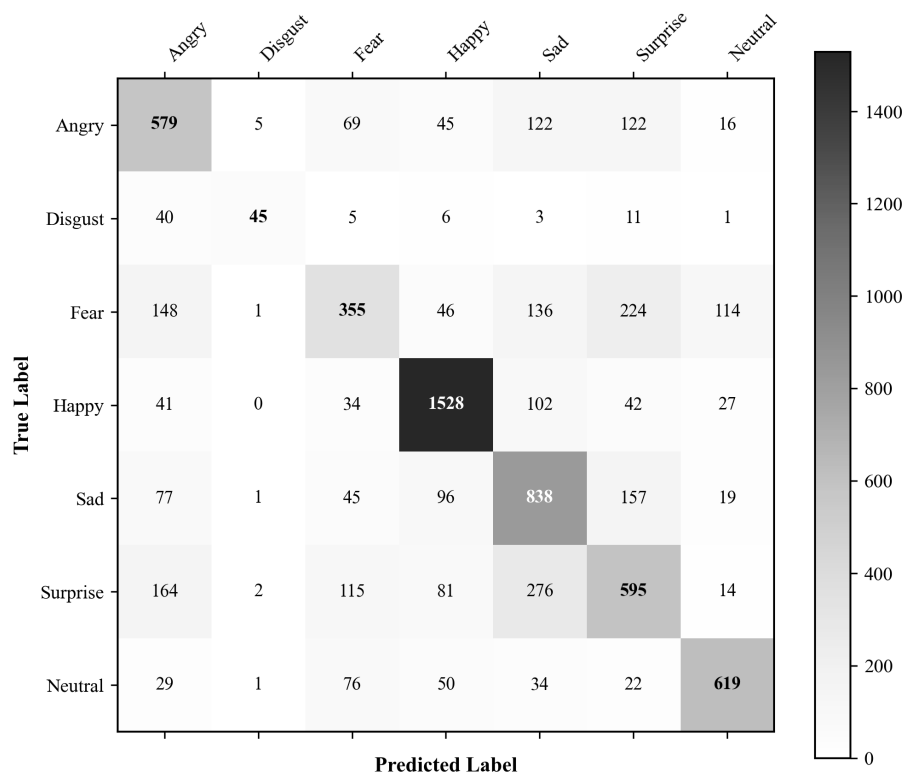


Figure 5.1: Grayscale Confusion Matrix Heatmap on 7,178 Test Images

Figure 5.2 shows the per-class F1-Score comparison, confirming that Happy and Neutral achieve the highest performance (84.28% and 75.44% respectively) while Fear achieves the lowest F1-Score (41.21%) due to high inter-class confusion with Surprise and Sad.

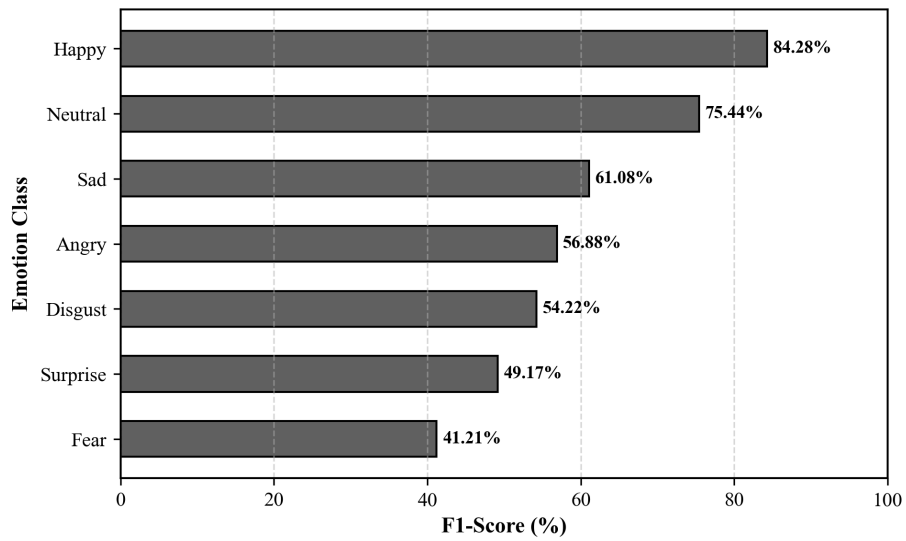


Figure 5.2: Per-Class F1-Score Performance Comparison

5.5 Real-Time Latency and FPS Performance

During live webcam deployment on the Windows environment (Intel Core i5, 16GB RAM, NVIDIA GeForce RTX 2050), the system sustains approximately 28–30 FPS processing throughput. MediaPipe face detection latency is approximately 3–5 ms per frame. CNN inference latency is approximately 12–18 ms per face crop on CPU (TensorFlow Intel-optimised). The 10-frame temporal smoothing buffer introduces a negligible latency of approximately 333 ms ($10 \text{ frames} \times 1/30\text{s}$) for stable emotion display.

Chapter 6: Future Work

While the proposed system achieves performance approaching the human baseline, several limitations and enhancement opportunities have been identified:

- **Face-Specific Pretrained Encoders:** Replacing ImageNet-pretrained backbones with encoders pretrained on large-scale face datasets such as VGGFace2 (3.31M images) or AffectNet (1M images) would provide substantially more semantically relevant feature representations for facial Action Units.
- **Vision Transformer (ViT) Architectures:** Self-attention mechanisms in Vision Transformers can capture global facial structure relationships and micro-expression patterns that local convolutional filters miss.
- **Mixup and CutMix Data Augmentation:** Generating synthetic training samples by linearly interpolating between minority and majority class face images would directly address class imbalance at the data level.
- **Multimodal Emotion AI:** Fusing facial expression features with speech audio emotion analysis (mel-spectrogram CNNs or wav2vec models) and physiological signals (heart rate, EEG) would improve robustness.
- **AffectNet Pre-training:** Fine-tuning on AffectNet (8 emotion classes, natural face images) before FER2013 fine-tuning would provide superior weight initialisation, particularly for minority classes.

Chapter 7: Conclusions

This project successfully designed, implemented, evaluated, and deployed a complete real-time facial emotion recognition system addressing the dual challenges of dataset class imbalance and live video inference performance.

The core contributions of this work are: (1) formulation of a Categorical Focal Loss ($\gamma = 2.0$) with Label Smoothing ($\epsilon = 0.1$) specifically addressing FER2013's severe class imbalance; (2) an upgraded 4-Stage Deep ConvNet architecture with double convolutional layers, Batch Normalization, L2 regularisation, and Global Average Pooling achieving 63.51% test accuracy; (3) integration of Google MediaPipe BlazeFace for robust real-time face detection; and (4) a 10-frame temporal smoothing buffer eliminating prediction flickering during live deployment.

The system achieves 63.51% test accuracy and 62.91% weighted F1-score on 7,178 FER2013 test images, closely approaching the human annotation baseline of approximately 65%. The system meets all stated project objectives and provides a robust, validated, and deployable framework for real-time Affective Computing applications, with clear pathways for future enhancement through face-specific transfer learning and multimodal fusion.

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Appendices

Appendix C: Source Code Repository

The complete source code for this project is available on GitHub at: **https://github.com/Max-i26/emotion_detection**

Key source files include:

- `train.py` — Upgraded 4-Stage ConvNet training pipeline with Focal Loss
- `live_detect_pro.py` — Primary real-time MediaPipe + CNN emotion detector
- `evaluate_all.py` — Comprehensive test set evaluation with classification report
- `generate_report_assets.py` — Figure generation scripts for this report
- `run_webcam.bat` — Single-click Windows launcher

Appendix D: Performance Metrics JSON

Full evaluation results including per-class precision, recall, F1-score, support values, and the raw confusion matrix are saved in machine-readable JSON format in `performance_metrics.json` within the repository root.